

Manual

Material Buffers and Inventories MPL-MPB 8.2

Version 1.0.23049

Last changed on: 02.09.2020

Copyright

©Copyright 2020 All rights reserved.

SAP® and R/3® are registered trademarks of SAP AG.

WINDOWS® is a registered trademark of Microsoft Corporation.

MPDV® and HYDRA® are registered trademarks of MPDV Mikrolab GmbH.

ORACLE® is a registered trademark of ORACLE Corporation, California, USA.

Copying and distribution of this documentation or any part thereof, for any purpose or in any form, is prohibited without prior written permission from MPDV Mikrolab GmbH.

The information contained in this documentation is subject to change without prior notice.

Contents

1	Overview of Material Buffers and Inventories	4
2	Applications Available in MOC	5
3	Stock Monitoring	6
4	Requirements Overview	9
5	Stock Overview	11
6	TPU Stock Overview	14

1 Overview of Material Buffers and Inventories

Purpose

The material buffer and inventories function package allows for material stock to be managed and evaluated.

Implementation Considerations

The function package "material buffers and inventories" is used if you

- use the function packages "material and inventory management", "batch data management" and/or "batch processing" and
 - would like to enhance them by managing storage locations / storage bins;
 - would like to record which anonymous, logistical containers include or transport the material;
- would like the system to calculate the consumption of posted input materials (retrograde);
- would like to enter the consumption of posted input materials explicitly.

Integration

- Input and output material buffers can be defined at the object "machine/workplace".

Features

- Management of buffer inventories
 - Management of material stock relating to material buffers or transport units
- Monitoring options for inventories
 - Monitoring of material stock relating to material buffers or transport units
- Management and monitoring of material requirements relating to existing inventories

2 Applications Available in MOC

Application	Documentation	Transaction code	Function authorization
Stock monitoring	MOC_StockMonitoring.docx	stomo	stomo
Requirements overview	MOC_RequirementsOverview.docx	reqov	reqov
Stock overview	MOC_StockOverview.docx	stov	stov
Stock overview TPU	MOC_StockOverviewHU.docx	stovhu	stovhu

3 Stock Monitoring

Summary

Menu	Material management → Inventory management → Stock monitoring
Transaction code	stomo
Function authorization	stomo

Usage

Stock monitoring allows to display those materials (batches), for which the [inventory limits](#) are exceeded or not reached in a [material buffer](#).

Integration

Since these material buffers can have different sizes and since the definition of when such a material buffer is full depends on the size of the material, the definition of the inventory limits to be monitored is referred to a material and to a [material buffer](#). This creates the possibility to monitor only critical materials and/or only certain material buffers.

All materials per material buffer that match the entered selection criteria and for which [inventory limits](#) were defined for the respective material buffer will be shown in this list that updates cyclically. To do so, the existing batches will be cumulated to the material number and be displayed. To further limit this, only batches of a specific status can be used.

Requirements

The following requirements must be met to display materials in stock monitoring:

1. In the configuration (menu: *Master data > Material > Inventory limits*) you must have defined corresponding inventory limits for the [combination of material and material buffer](#).
2. The [material buffer](#) (menu: *Master data > Material > Material buffer*) must not be a hierarchical buffer. In addition the indicator "include in stock" must be set.

Selection criteria

The following selection criteria are available in the application:

Material buffer

This selection criterion refers to the material buffer of the material. All materials with the selected [material buffer](#) will be shown.

Material

Any stock with the selected material will be shown.

Batch status

This selection criterion refers to the batch status of the stocks. All materials with the selected batch status will be shown.

If several selection criteria are used, the intersection of the selection criteria will be shown.

Field descriptions

Fill level

Symbol	Symbol color	Meaning	Meaning related to the configured inventory limits
	green	Stock o.k.	Stock between min. warning stock level and max. warning stock level.
	yellow	Stock below warning limit	Stock between min. warning stock level and minimum stock.
	red	Stock below alarm limit	Stock between min. alarm quantity and min. warning stock level
	red	Stock below minimum	Stock below minimum stock
	yellow	Stock above warning limit	Stock between max. warning stock level and max. alarm quantity.
	red	Stock above alarm limit	Stock between max. alarm quantity and max. stock level
	red	Stock above maximum	Stock above max. stock level

Material type

Assigned material type of the material

Material

Material number

Material designation

Designation of the material

Quantity

Stock quantity of the material

Unit

Quantity unit

Material buffer

Assigned material buffer of the material

Designation

Designation of the material buffer

Company

Assigned company to the material buffer

Department

Assigned department to the material buffer

Area

Assigned area to the material buffer

Cost center

Assigned cost center to the material buffer

Storage location

Assigned storage location to the material buffer

Type

Type of the material buffer

Min. stock quantity

Defined min. stock quantity of the material in this material buffer

Lower alarm limit (min. alarm quantity)

Defined min. alarm quantity of the material in this material buffer

Lower warning limit (min. warning stock level)

Defined lower warning stock level of the material in this material buffer

Upper warning limit (max. warning stock level)

Defined upper warning stock level of the material in this material buffer

Upper alarm limit (max. alarm quantity)

Defined upper alarm quantity of the material in this material buffer

Max. stock level

Defined max. stock level of the material in this material buffer

4 Requirements Overview

Summary

Menu	Material management → Inventory management → Requirements overview
Transaction code	reqov
Function authorization	reqov

Usage

The function “material requirements” provides an overview of the materials required at machines. It is particular to this function that material requirements are exactly determined by day taking the shift calendar into account. Consequently, material requirements are distributed according to the run time of OP(s). “By day” means here “by shift”, i.e. the day starts with the beginning of shift 1 and ends with the end of the last shift (e.g. shift 3).

Requirements

The PPS system transfers the materials required for the production of an operation and, as a result, they are available in the component list of the OP within the system.

Please note:

The determination of material requirements depends, among others, on the planned start date of an operation. For this reason, the planned start date of an operation needs to be transferred correctly by the PPS system or planned in shop floor scheduling. If, however, the order sequencing function is used this planned date is used to define the order of processing (internal algorithm). In this case, the planned start date can no longer be derived.

Selection criteria

The result of overlapping selection criteria is displayed if several selection criteria are in use.

The application provides the following selection criteria:

Material

Shows all materials assigned to the selected material.

Workplace

Shows all materials assigned to the selected workplace.

MES order number

Shows all materials assigned to the selected MES order number.

Material type

This selection criterion refers to the material type of the material. All materials assigned to the selected material type are shown.

Planned start

Shows all materials assigned to the selected planned start.

Consider pool of groups

Defines whether or not the pool of groups is to be taken into account for selection.

5 Stock Overview

Summary

Menu	Material management→ Inventory management→ Stock overview
Transaction code	stov
Function authorization	stov

Usage

The stock overview is an evaluation of the material management. Here, the employees of the warehouse, dispatch and production supply or logistics departments can inspect the current buffer stock and/or stock in production.

Integration

This evaluation shows the batch and material stock within a [material buffer](#) cumulated for each material and each batch status within the material, i.e. the material number. Such batches with a quantity of 0 and batches that are not assigned to a [material buffer](#) will not be taken into account.

Selection criteria

The result of overlapping selection criteria is displayed if several selection criteria are in use.

The following selection criteria are available in the application:

General

Batch number

(Externally used) batch number that is to be selected.

Material

All materials with the selected material number will be shown.

Internal batch number

Internal batch number that is unique within the system and that is to be selected.

Material buffer

This selection criterion refers to the [material buffer](#) of the batch. All batches with the selected material buffer will be used for the evaluation.

Material type

This selection criterion refers to the material type of the material. All materials with the selected material type will be shown.

Batch status

This selection criterion refers to the batch status of the material. All materials with the selected batch status will be shown.

Attributes**Attributes 1 - 10**

Selecting an attribute as a selection criterion displays all inventories of batches that currently have this identifier.

Batch attributes**Batch attributes**

Selecting a batch attribute as a selection criterion displays all inventories of batches for the material type of which this batch attribute has been configured and that are currently assigned this identifier.

Alternative batch numbers**Alternative batch number 1-20**

Selecting an alternative batch number as a selection criterion displays all inventories of batches that currently have this identifier.

Field description

Primarily, the evaluation shows the current information on a batch as well as information on the material buffer concerned.

Information on batch

The information per [batch](#) is the same as the one presented in the [batch data overview](#).

Information on material buffer

In the stock overview the information on the [material buffer](#) will be shown per batch. This is primarily:

Material buffer

Material buffer of the batch

Designation

Designation of the material buffer

Storage location

Assigned storage location of the material buffer

Type

Type of the material buffer

6 TPU Stock Overview

Summary

Menu	Material management→ Inventory management→ TPU stock overview
Transaction code	stovhu
Function authorization	stovhu

Usage

The transport units stock overview is used to show the quantities of materials in the individual transport units. This can be used to determine which quantities of which material and which batches are included in which transport unit and how the transport units are currently assigned.

Integration

A pre-selection of material, transport unit and of other criteria will be shown as cumulative display of the batch/ material stock per transport unit. The display will be accumulated for each material, transport unit and batch status. Batches with a remaining quantity of 0 will not be accounted for.

Only those materials will be accounted for that are assigned to those material types, for which the "inventory management" indicator is not set to No.

Moreover, only those transport units will be accounted for, for which the "inventory management" indicator is set to "Y".

Selection parameters

The result of overlapping selection criteria is displayed if several selection criteria are in use.

The following selection criteria are available in the application:

General

Batch number

All materials with the selected batch number will be shown.

Material

All materials with the selected material will be shown.

Material type

This selection criterion refers to the material type of the material. All materials with the selected material type will be shown.

Batch status

This selection criterion refers to the batch status of the material. All materials with the selected batch status will be shown.

Transport unit

All materials with the selected transport unit will be shown.

Attributes**Attributes 1 - 10**

Selecting an attribute as a selection criterion displays all inventories of batches that currently have this identifier.

Batch attributes**Batch attributes**

Selecting a batch attribute as a selection criterion displays all inventories of batches for the material type of which this batch attribute has been configured and that are currently assigned this identifier.

Alternative batch numbers**Alternative batch number 1-20**

Selecting an alternative batch number as a selection criterion displays all inventories of batches that currently have this identifier.

Field description

Calculation of the values in the "inventory" category:

Number of batches

Quantity of batches (in line with the stock indicator set to the material type)

Quantity

Sum of yield + scrap

Unit

Unit from the configuration table TPU – material type

Yield

Quantity identified as yield that is in fact included in the "number of batches"

Scrap

Quantity identified as scrap that is in fact included in the "Number of batches"

Calc. Quantity

Quantity of material that would fit into the currently assigned transport units:

Quantity from the configuration TPU material type for this TPU x number of batches

Calc. Quantity of TPU

Quantity of TPU that would be necessary for the total quantity:

Quantity/ (quantity from the configuration TPU material type for this TPU)

The result will be rounded up to the next integer.

Batches that are not assigned to a transport unit will **not** be accounted for in this evaluation.